

Assignment I - PHYS 708

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Problem 1

(a) Obtain an expression for the threshold kinetic energy of an incident proton to produce a neutral pion in a proton-proton interaction:

$$p + p \rightarrow p + p + \pi^0.$$

(b) Find the numerical value of E_k for the interaction.

Solution:

(a) We solve this problem by finding the invariants s_{LAB} and s_{CM} for both the Lab and CM frames. Note: We take the speed of light $c = 1$ to ease calculations.

Setup

Lab frame: p_1 incident proton with total energy E and kinetic energy $E_k = E - m_p$, p_2 proton at rest

$$p_1 = (E, \mathbf{p}), \quad p_2 = (m_p, \mathbf{0}).$$

So

$$s_{LAB} = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = m_p^2 + m_p^2 + 2(E m_p) = 2m_p^2 + 2m_p E.$$

CM frame: All particles are at rest. Therefore, s_{CM} is the square sum of the rest energies of the particles.

$$s_{CM} = (2m_p + m_{\pi^0})^2.$$

Equate the two expressions for s :

$$2m_p^2 + 2m_p E = (2m_p + m_{\pi^0})^2.$$

Solve for E :

$$E = \frac{(2m_p + m_{\pi^0})^2 - 2m_p^2}{2m_p} = \frac{2m_p^2 + 4m_p m_{\pi^0} + m_{\pi^0}^2}{2m_p} = m_p + 2m_{\pi^0} + \frac{m_{\pi^0}^2}{2m_p}.$$

The threshold kinetic energy is $E_{th} = E - m_p$:

$$E_{th} = 2m_{\pi^0} + \frac{m_{\pi^0}^2}{2m_p} = \frac{m_{\pi^0}}{2m_p} (4m_p + m_{\pi^0}).$$

(b) Using $m_{\pi^0} = 135$ MeV, $m_p = 938$ MeV:

$$E_{th} = \frac{135}{2 \cdot 938} (4 \cdot 938 + 135) \text{ MeV} \approx 279.7 \text{ MeV} \approx 280 \text{ MeV}.$$

Problem 2

(a) Show that the threshold Lorentz factor of an incident proton to produce an e^-e^+ pair in a proton-photon interaction

$$p + \gamma \rightarrow p + e^- + e^+$$

is given by $\gamma_{th} \approx \frac{m_e}{\epsilon}$, where m_e is the rest mass energy of the electron and ϵ is the target photon energy. Consider the collision to be head-on.

(b) Calculate the threshold total energy of the incident proton for an interaction with a cosmic microwave background photon.

Solution:

(a)

Setup

Lab frame:

p_1 incident proton with total energy $E = \gamma m_p$ and momentum $p = \gamma m_p v$,

p_{photon} is a photon with energy $E = \epsilon$ and momentum $p = -E = -\epsilon$

We assume the head-on collision is on the x-axis to simplify the momentum vector.

$$p_1 = (\gamma m_p, \gamma m_p v, 0, 0), \quad p_{photon} = (\epsilon, -\epsilon, 0, 0).$$

So

$$s_{LAB} = (p_1 + p_{photon})^2 = p_1^2 + p_{photon}^2 + 2p_1 \cdot p_{photon} = m_p^2 + 0^2 + 2(\gamma m_p \epsilon - (-\gamma m_p v \epsilon)) = m_p^2 + 2\gamma m_p \epsilon (1 + v).$$

CM frame: All particles are at rest. Therefore, s_{CM} is the square sum of the rest energies of the particles.

$$s_{CM} = (m_p + 2m_e)^2.$$

If the proton is **ultrarelativistic**, $v \approx 1$. We equate the two expressions for s :

$$m_p^2 + 4\gamma m_p \epsilon = (m_p + 2m_e)^2.$$

Solve for γ :

$$\gamma_{th} = \frac{(m_p + 2m_e)^2 - m_p^2}{4m_p \epsilon} = \frac{(2m_p + 2m_e)(2m_e)}{4m_p \epsilon} = \frac{m_e}{\epsilon} \left(1 + \frac{m_e}{m_p} \right) \approx \frac{m_e}{\epsilon}.$$

since $m_p \gg m_e$.

(b) Using $m_p = 938 \text{ MeV}$, $m_e = 0.5 \text{ MeV}$, $\epsilon = 7 \times 10^{-4} \text{ eV}$:

$$E_{th} = \gamma_{th} m_p = \frac{m_e \cdot m_p}{\epsilon} = \frac{0.5 \times 938}{7 \times 10^{-4} \times 10^{-6}} \text{ MeV} = 67 \times 10^{10} \text{ MeV} \approx 7 \times 10^{17} \text{ eV}$$

Problem 3

(a) Show that the threshold Lorentz factor of an incident proton to produce a neutral pion in a proton-photon interaction

$$p + \gamma \rightarrow p + \pi^0$$

is given by

$$\gamma_{th} \approx \frac{m_{\pi^0}}{2\epsilon} \left(1 + \frac{m_{\pi^0}}{2m_p} \right),$$

where m_{π^0} is the rest mass energy of the neutral pion, ϵ is the target photon energy, and m_p is the proton rest mass energy. Consider the collision to be head-on.

(b) Obtain the value of the threshold total energy of the proton for an interaction on a cosmic microwave background photon.

Solution: (a) Setup

Lab frame:

p_1 incident proton with total energy $E = \gamma m_p$ and momentum $p = \gamma m_p v$,

p_{photon} is a photon with energy $E = \epsilon$ and momentum $p = -E = -\epsilon$

We assume the head-on collision is on the x-axis to simplify the momentum vector.

$$p_1 = (\gamma m_p, \gamma m_p v, 0, 0), \quad p_{photon} = (\epsilon, -\epsilon, 0, 0).$$

So

$$s_{LAB} = (p_1 + p_{photon})^2 = p_1^2 + p_{photon}^2 + 2p_1 \cdot p_{photon} = m_p^2 + 0^2 + 2(\gamma m_p \epsilon - (-\gamma m_p v \epsilon)) = m_p^2 + 2\gamma m_p \epsilon (1 + v).$$

CM frame: All particles are at rest. Therefore, s_{CM} is the square sum of the rest energies of the particles.

$$s_{CM} = (m_p + m_{\pi^0})^2.$$

If the proton is **ultrarelativistic**, $v \approx 1$. We equate the two expressions for s :

$$m_p^2 + 4\gamma m_p \epsilon = (m_p + m_{\pi^0})^2.$$

Solve for γ :

$$\gamma_{th} = \frac{(m_p + m_{\pi^0})^2 - m_p^2}{4m_p \epsilon} = \frac{(2m_p + m_{\pi^0})(m_{\pi^0})}{4m_p \epsilon} = \frac{m_{\pi^0}}{2\epsilon} \left(1 + \frac{m_{\pi^0}}{2m_p} \right).$$

since $m_p \gg m_e$.

(b) Using $m_{\pi^0} = 135 \text{ MeV}$, $m_p = 938 \text{ MeV}$:

$$E_{th} = \gamma_{th} m_p = \frac{m_{\pi^0}}{2\epsilon} \left(m_p + \frac{m_{\pi^0}}{2} \right) = \frac{135}{2 \cdot 7 \times 10^{-10}} \left(938 + \frac{135}{2} \right) \text{ MeV} \approx 9.70 \times 10^{13} \text{ MeV} \approx 9.70 \times 10^{19} \text{ eV}.$$